UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Reaction Time Analysis in a Stroop Task: Individual, Subgroup, and Group-Level Visualizations using MATLAB

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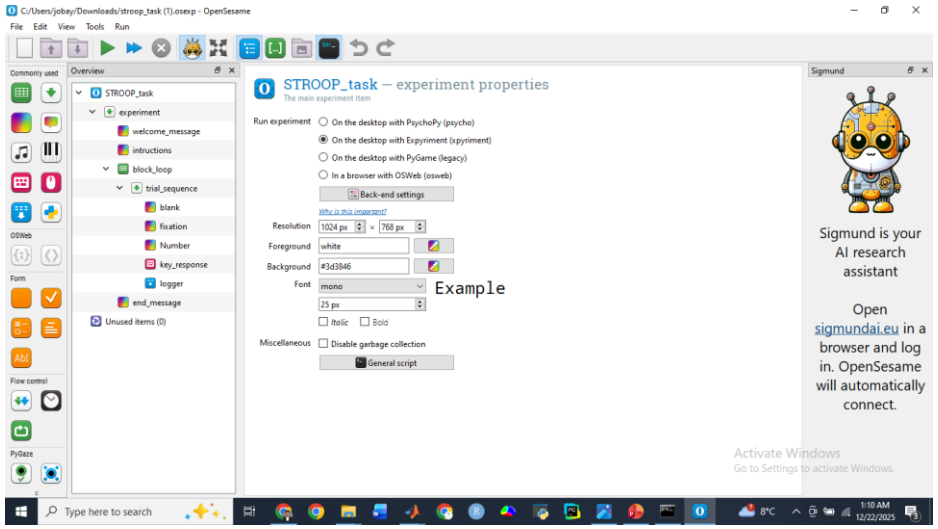
2110386
MATLAB AND COMPUTATIONAL COGNITION 2025-2026

The Stroop Task

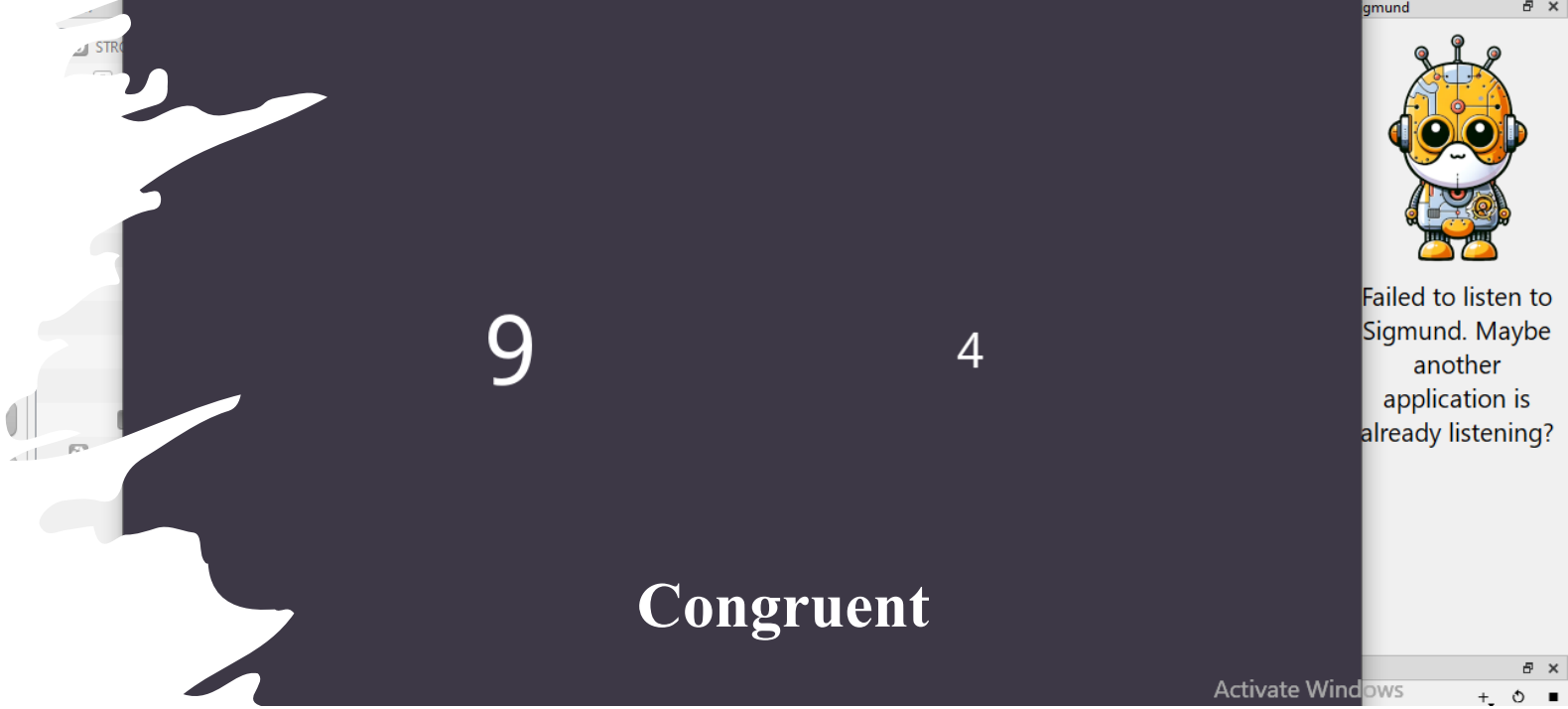
A **Stroop task** is a **psychological task** used to study **attention, cognitive control, and interference**.

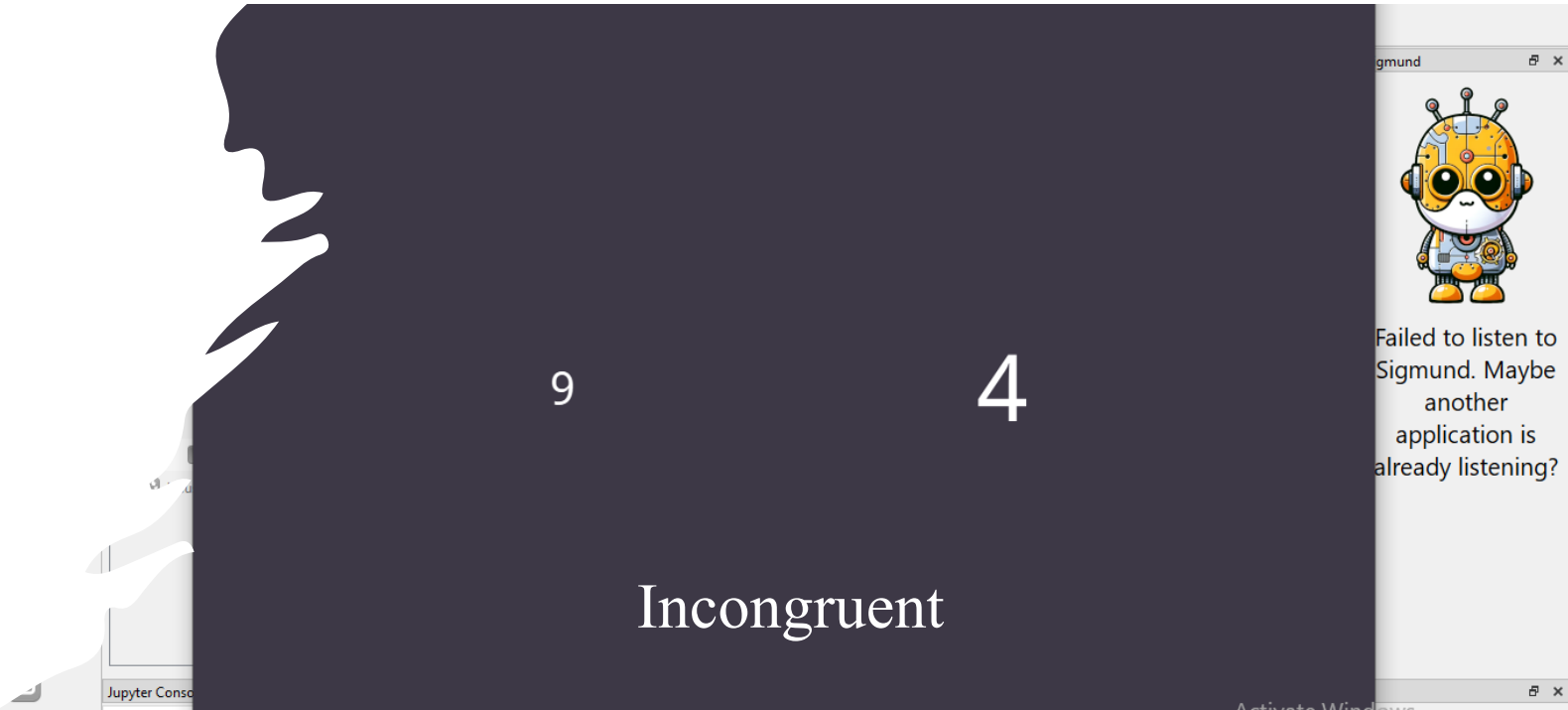
- To collect data I used a Numerical Stroop Task which we build using Opensesame in class.
- The independent variable in the task was the stimulus condition, which had three levels: neutral, congruent, and incongruent
.
- The dependent variable is Reaction time.



Goal: To assess differences in reaction time across neutral, congruent, and incongruent conditions.

Task Conditions





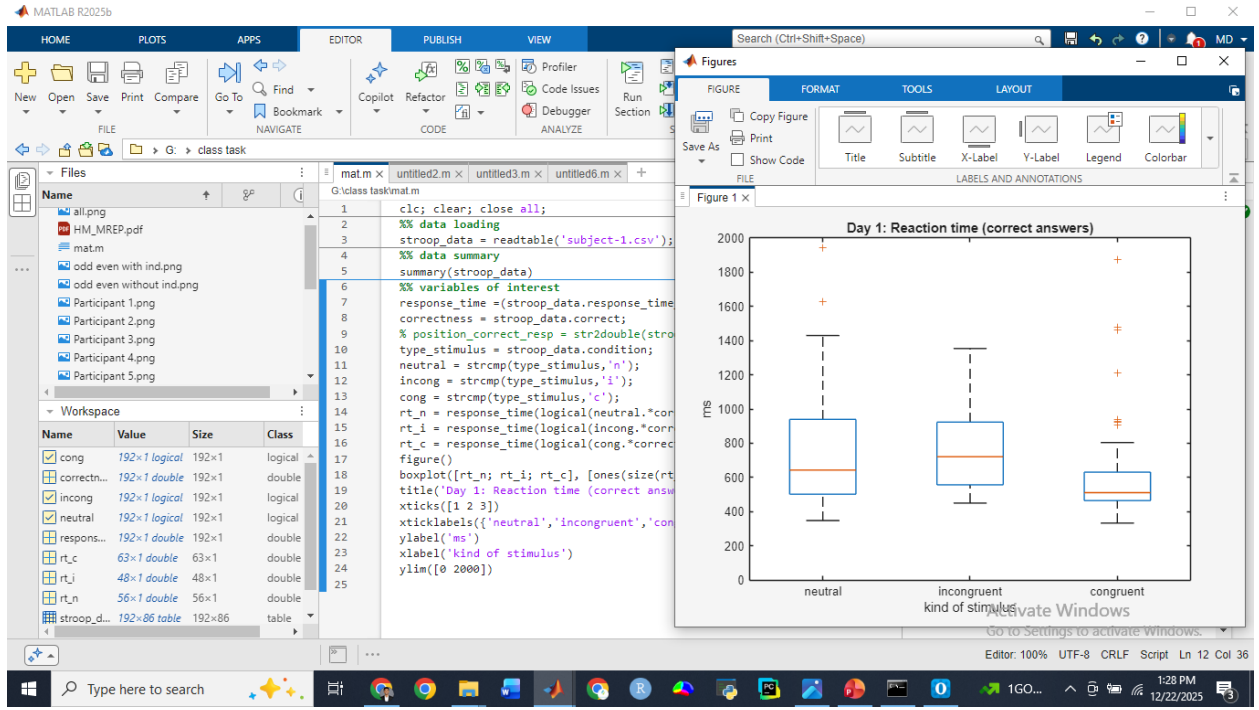
Dataset Overview

- In th experiment i have collected data from 6 paricipants.
- All of the were between 22-26 years old & university student.
  - Subject 1,3,5 are from bahelor program
  - Subject 2,4,6 are pursuing Masters

subject-1	12/15/2025 7:54 PM	Microsoft Excel C...	132 KB
subject-2	12/15/2025 8:10 PM	Microsoft Excel C...	132 KB
subject-3	12/15/2025 8:19 PM	Microsoft Excel C...	132 KB
subject-4	12/15/2025 8:41 PM	Microsoft Excel C...	134 KB
subject-5	12/15/2025 8:49 PM	Microsoft Excel C...	133 KB
subject-6	12/15/2025 11:21 PM	Microsoft Excel C...	131 KB

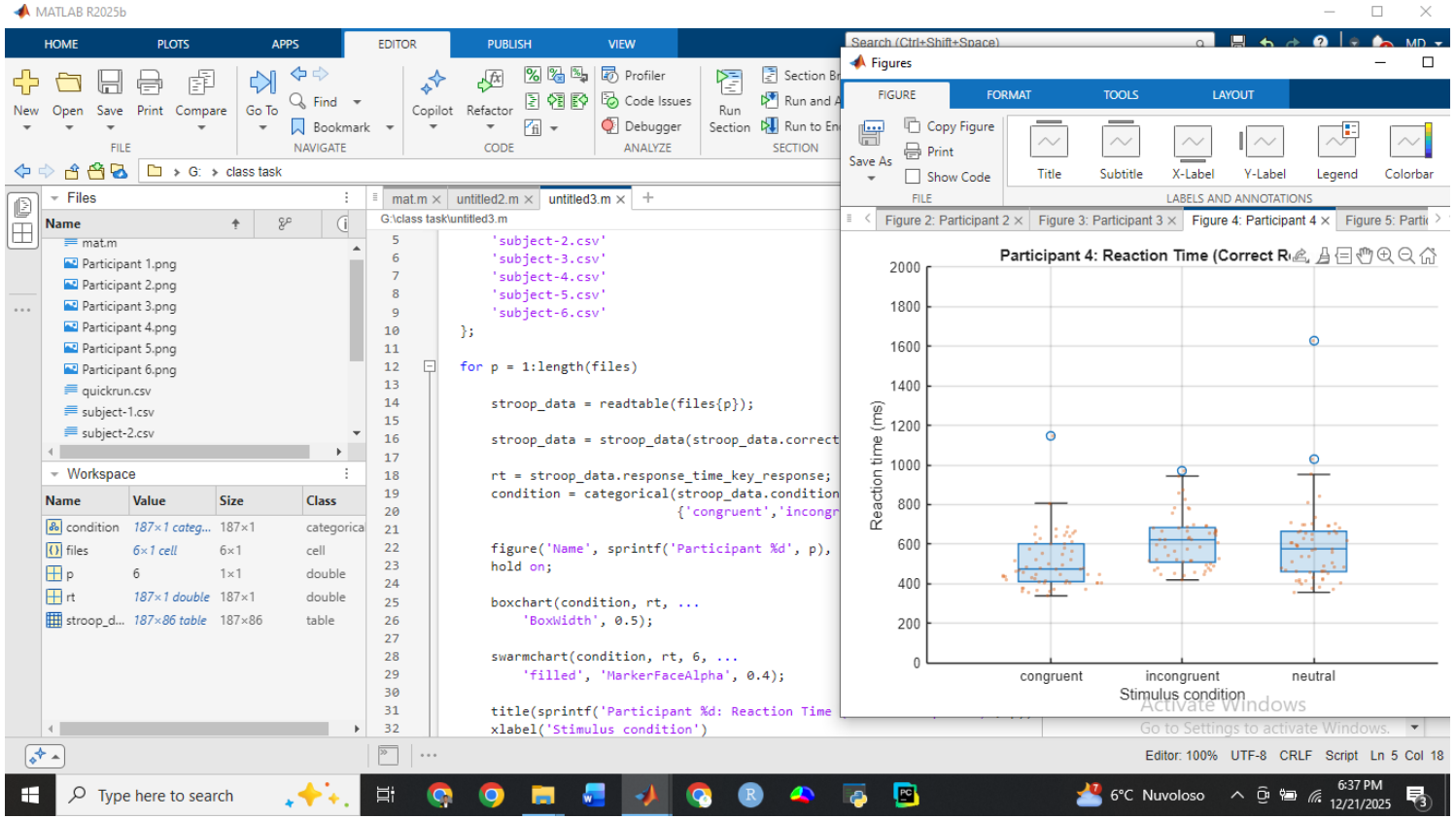
Analysis and Visualization using MATLAB

- Part: 1 - Individual Participant Visualization
- Part: 2 - Odd (Subject 1,3,5) vs Even (Subject 2,4,6) Participant split data
- Part: 3 - Pooled group level Visualization



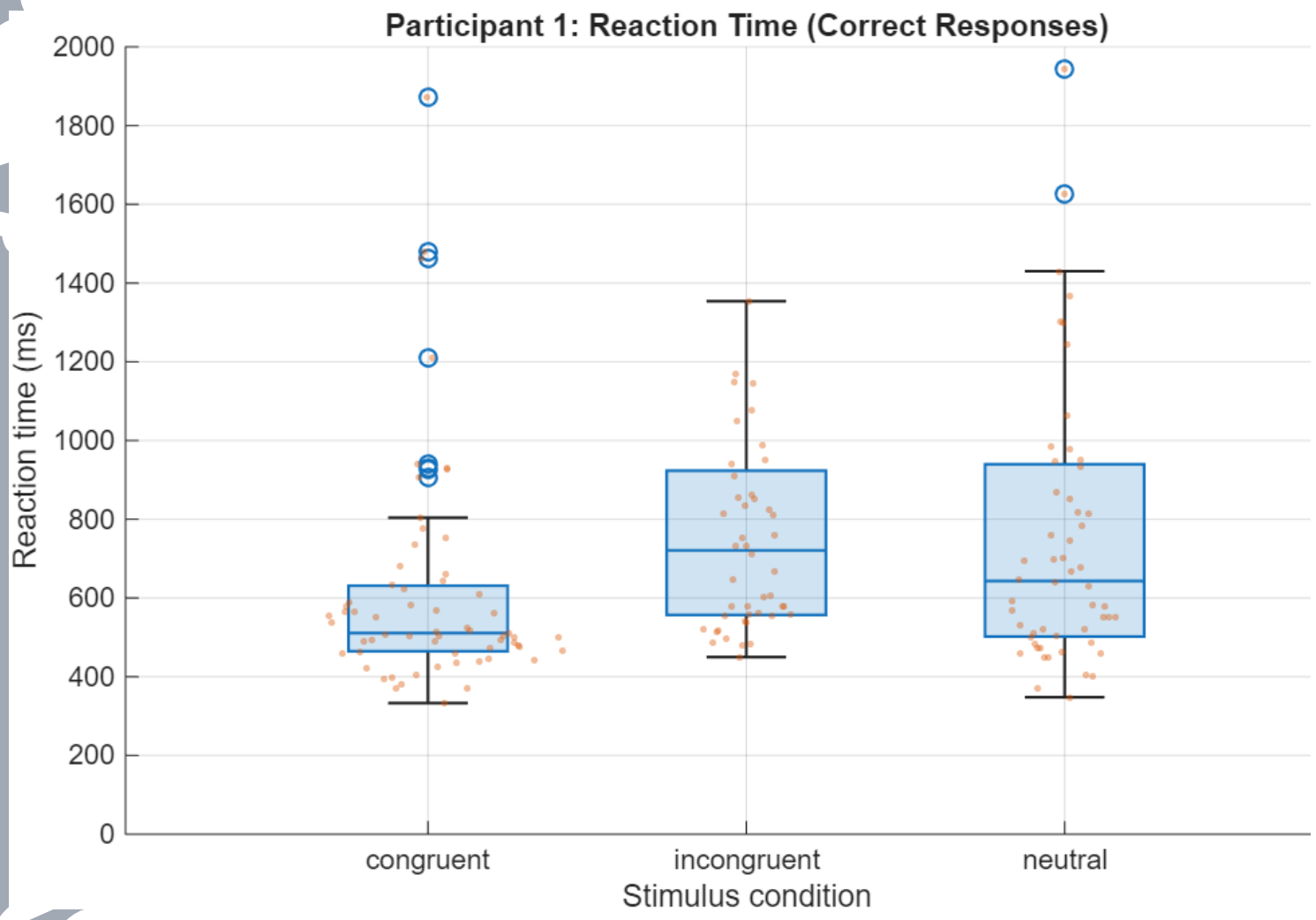
Part: 1 - Individual Participant Visualization

- I have used the MATLAB script provided in the class.
- Additionally, I have done a few modification of the script:
  - loading all the dataset together.
  - Using a “for loop” for each dataset and figure.



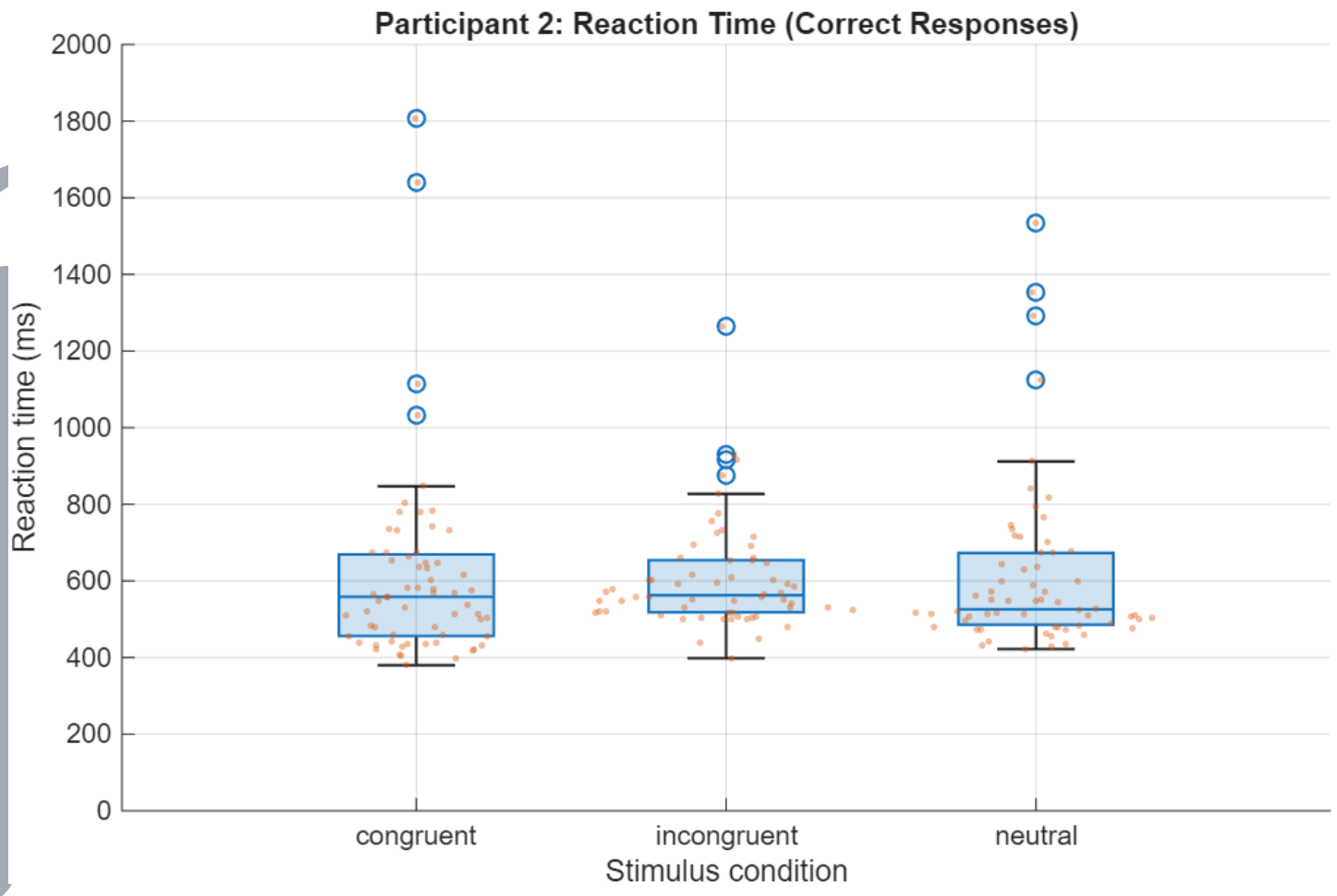
Part: 1 - Individual Participant Visualization

- Here, we can see that Participant 1 showed faster reaction times in the congruent condition (around 500ms), slower responses in the incongruent condition (around 700ms), and intermediate reaction times in the neutral condition (above 600ms), which suggests the presence of a Stroop interference effect, where conflicting stimulus information increases response time.



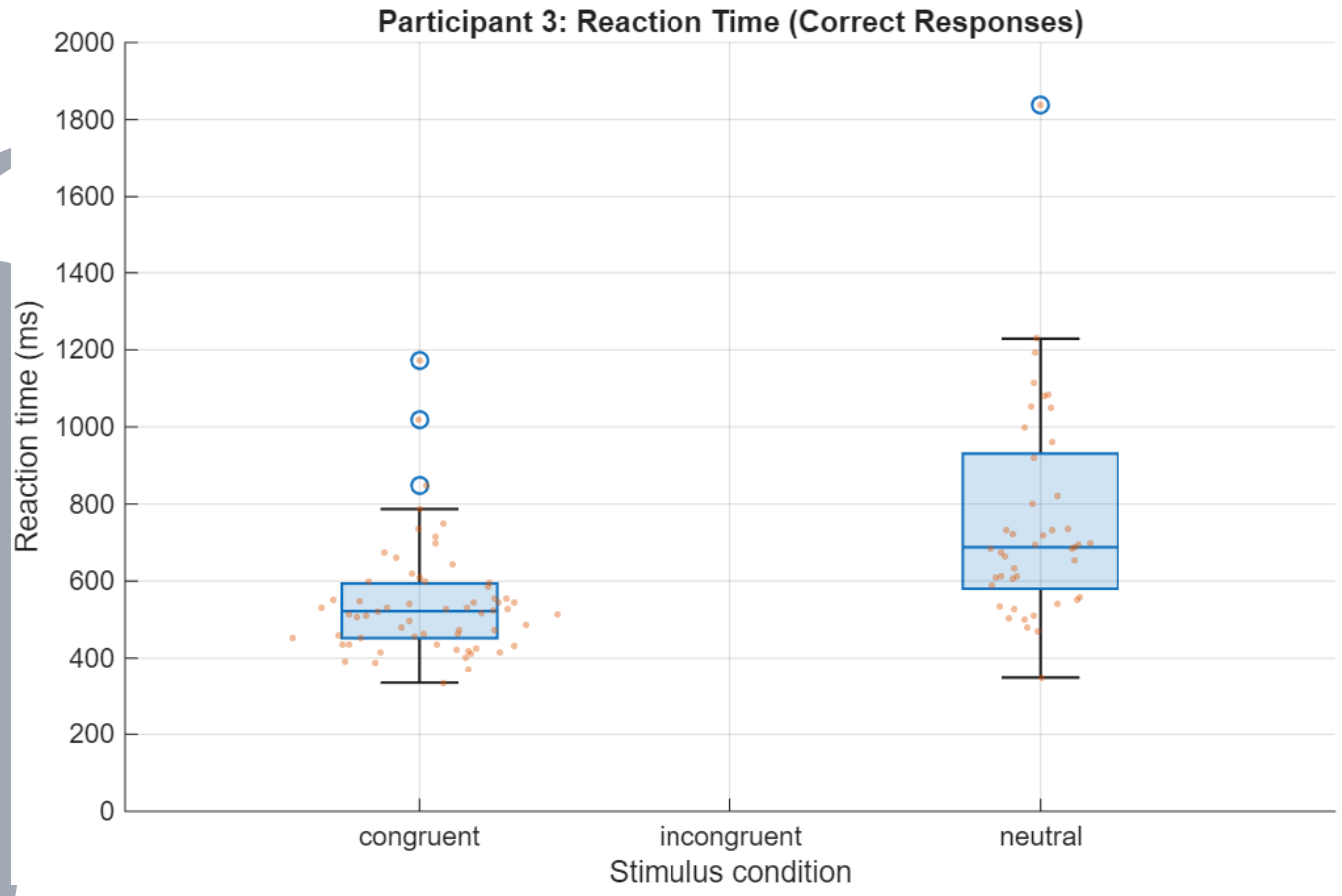
Part: 1 - Individual Participant Visualization

Here, we can see that Participant 2 showed not much of stroop effect in any condition .



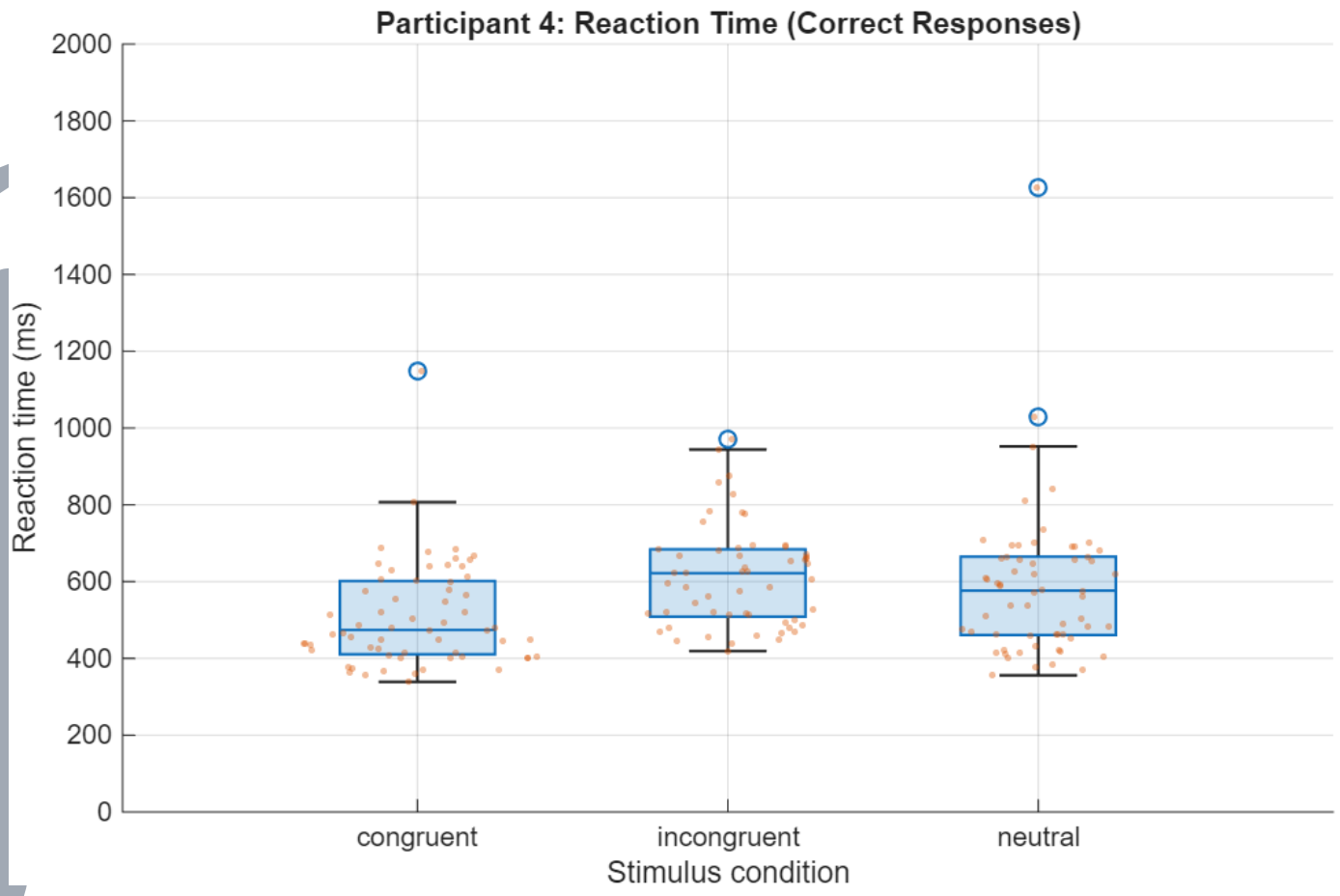
Part: 1 - Individual Participant Visualization

- Here, we can see that Participant 3 showing more faster response in congruent condition than neutral condition and there were no correct answer for incongruent tasks.



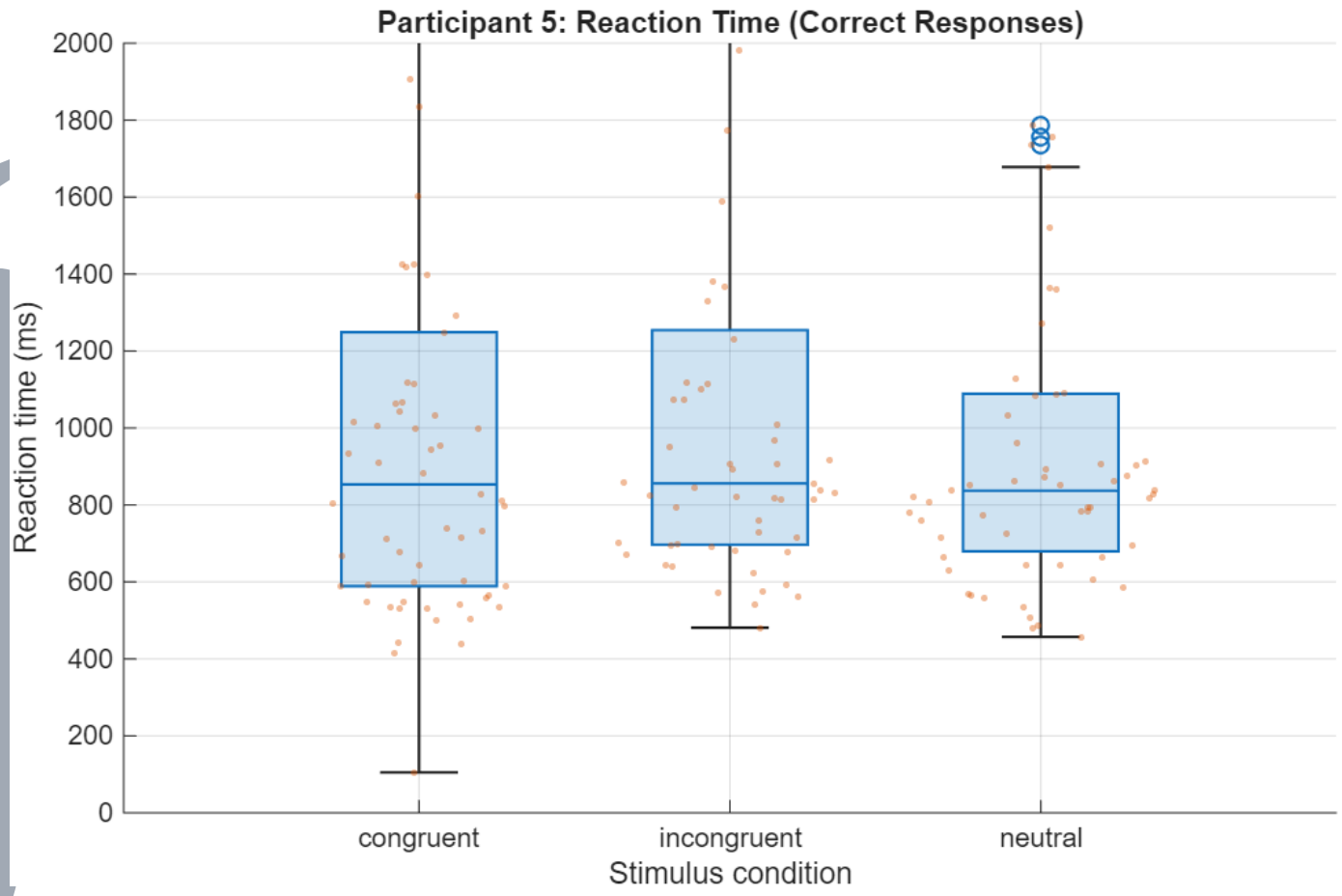
Part: 1 - Individual Participant Visualization

- Here, we can see that Participant 4 showed the stroop effect.

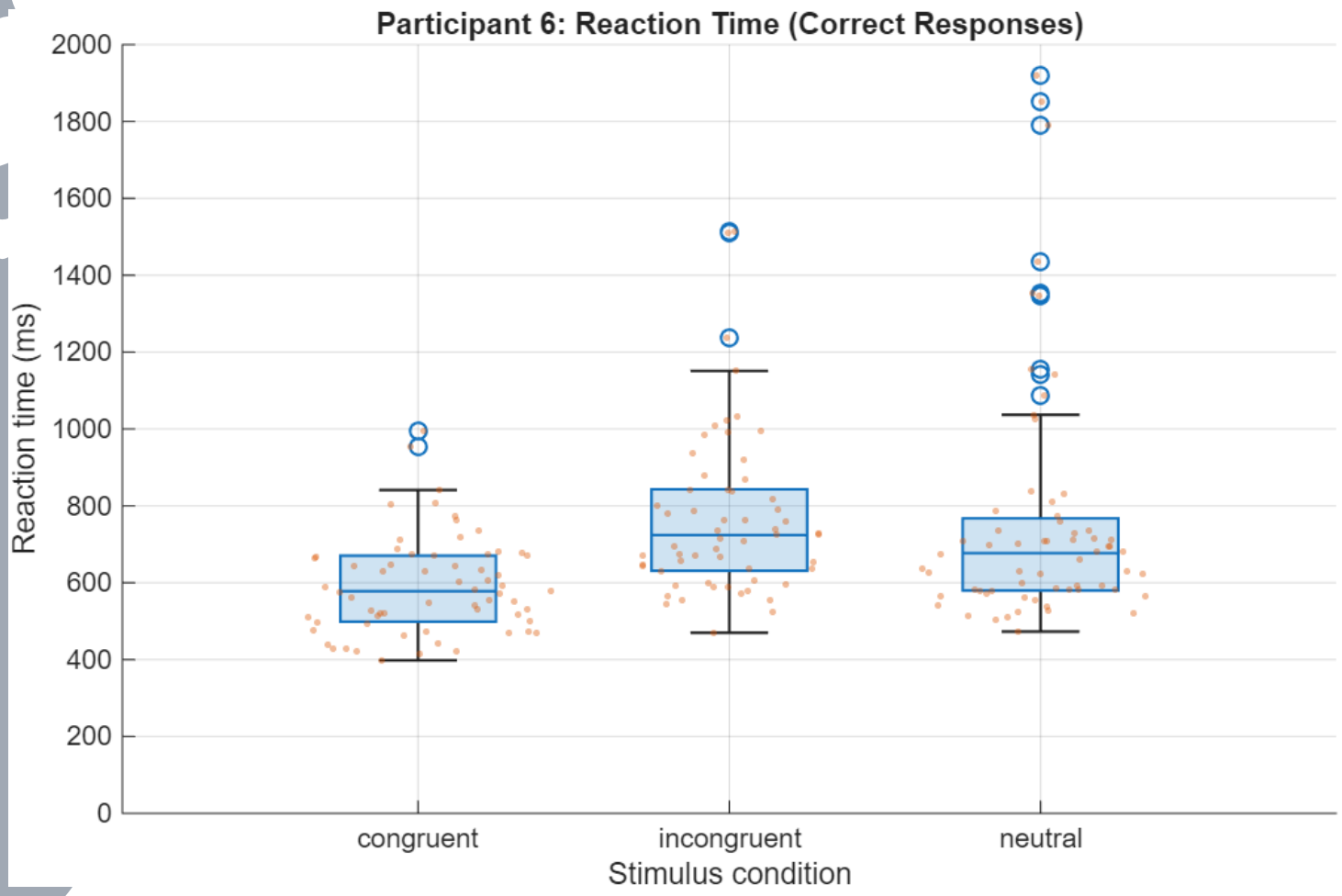


Part: 1 - Individual Participant Visualization

- Here, we can see that Participant 5 not showing any effect at all in all condition.



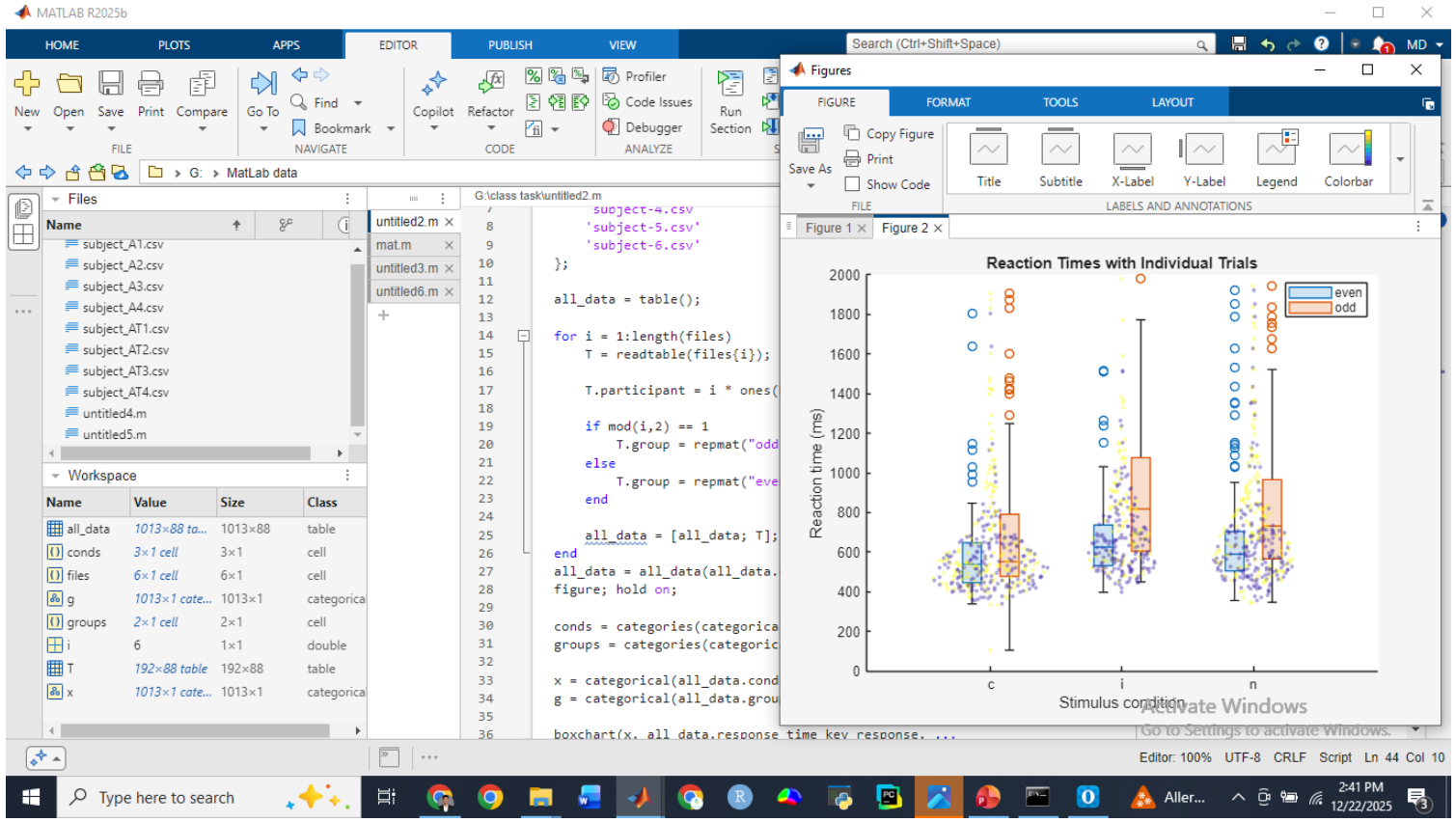
Part: 1 - Individual Participant Visualization



- Here, we can see that Participant 6 has the stroop effect in the incongruent ccondition task.

Part: 2 - Odd vs Even Participant split data

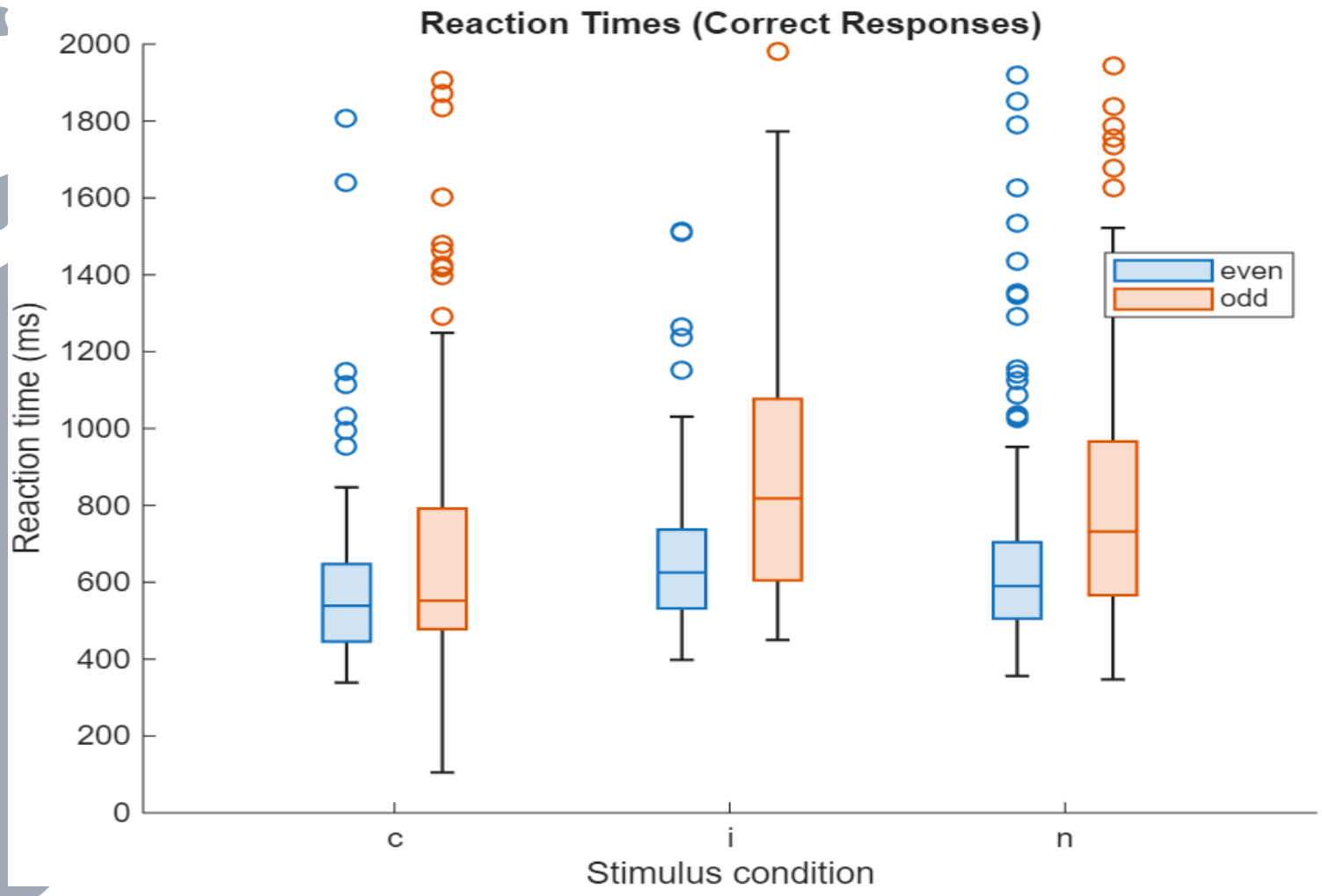
Here, I have used the same MATLAB script and then defined the dataset into two groups and plotted them in a figure to show the differences between those groups in different conditions.



Part: 2 - Odd vs Even Participant split data

- Here,
  - The odd Participants – Bachelor students
  - The even Participants – Masters students

- We can see that in incongruent and neutral conditions master student showed less stroop effect than bachelor students.



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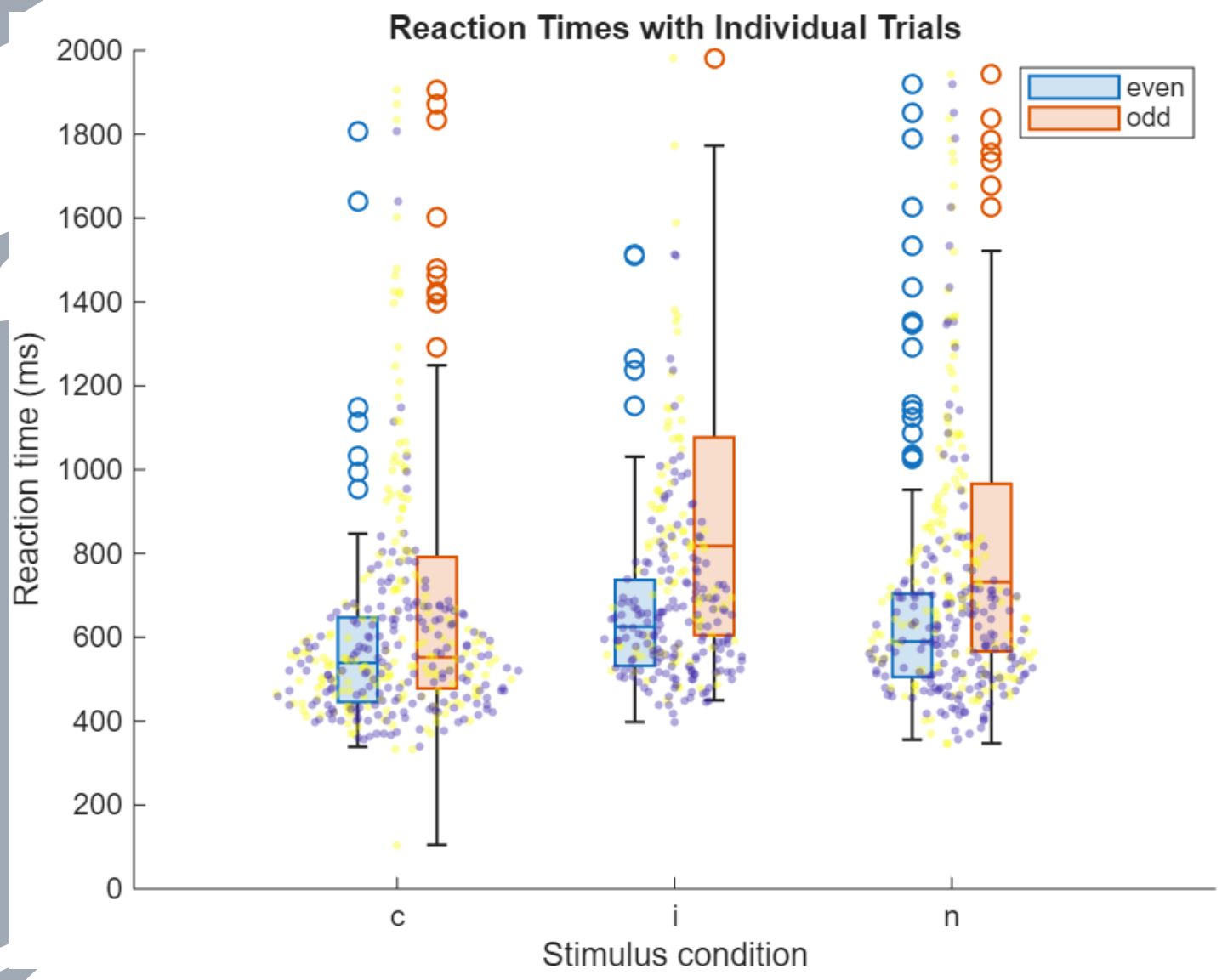


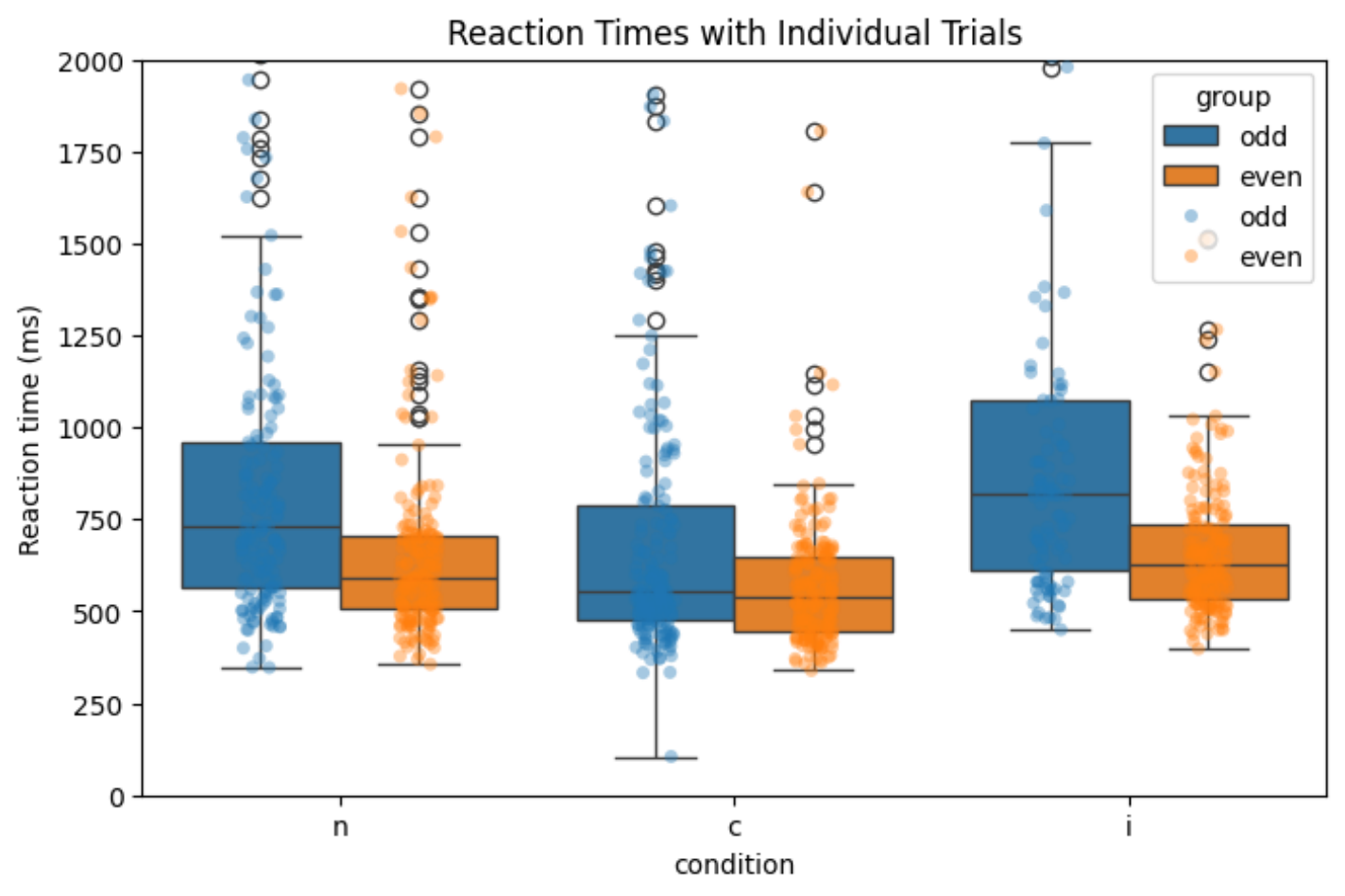
Part: 2 - Odd vs Even Participant split data

Here,

The odd Participants – Bachelor students

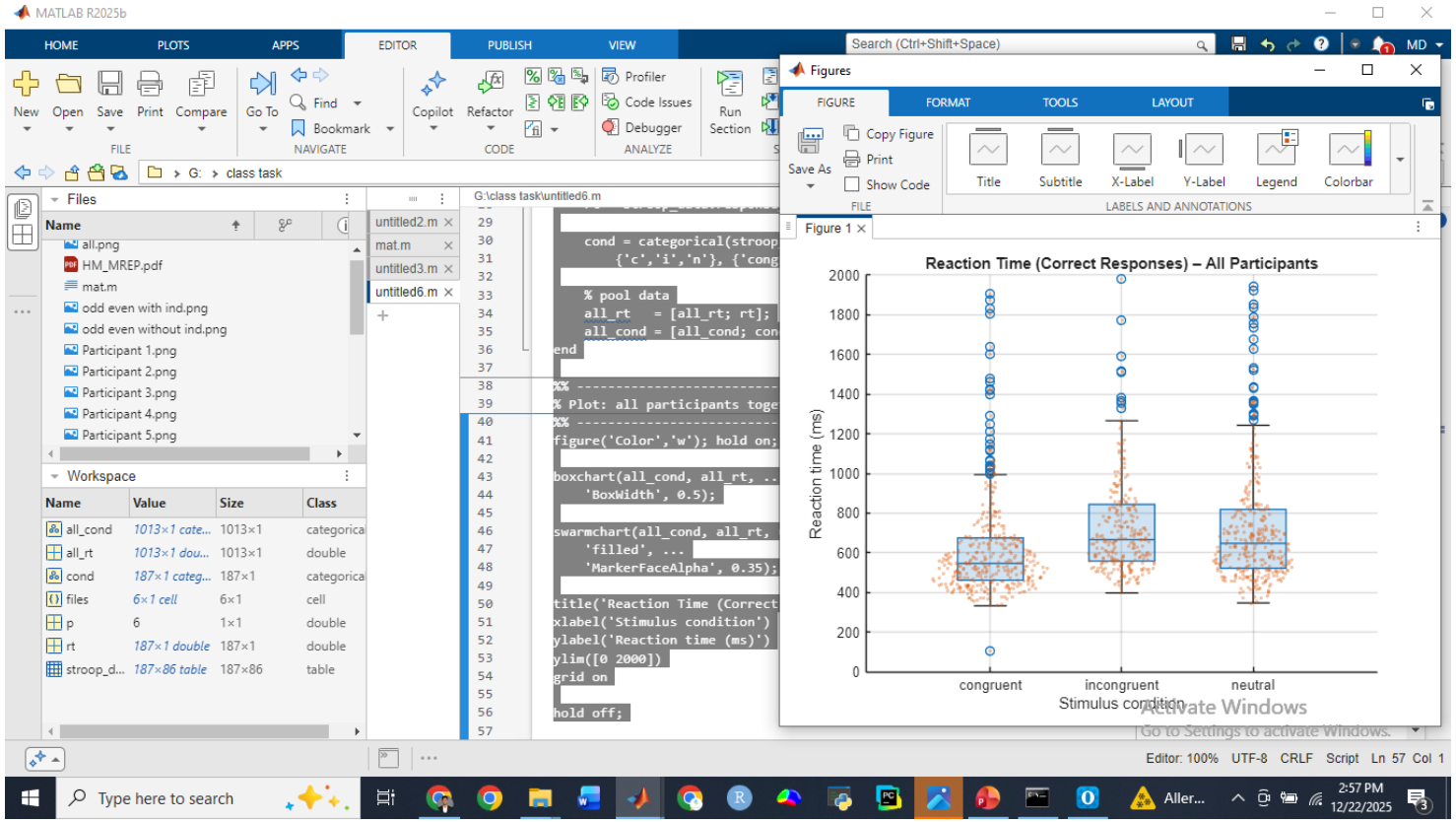
The even Participants – Masters students

I did it with MATPLOTLIB, and put it here just the color looked nice.



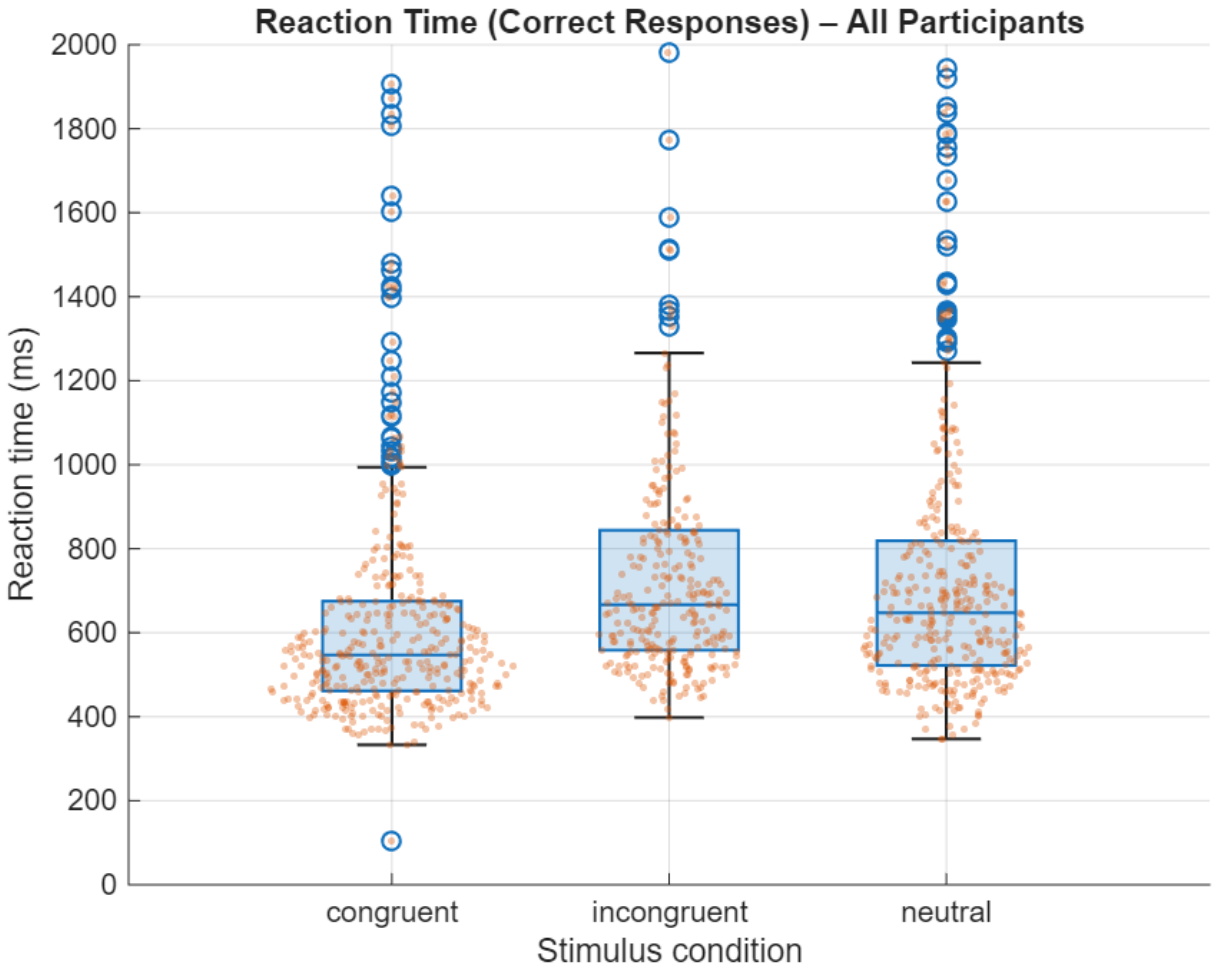
Part: 3 - Pooled group level Visualization

Here, I have used the same MATLAB script and used all the dataset into one figure.



Part: 3 - Pooled group level Visualization

Here, we can see all on average in all datasets incongruent reaction time was higher, thus explaning the stroop effect.



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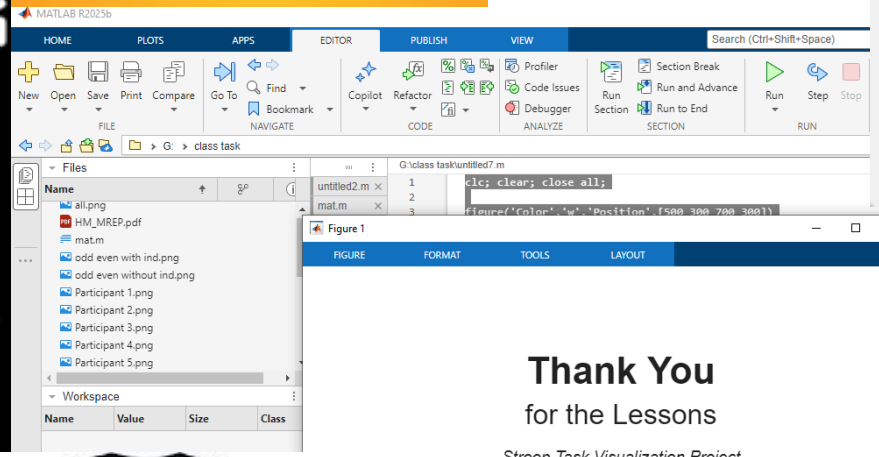

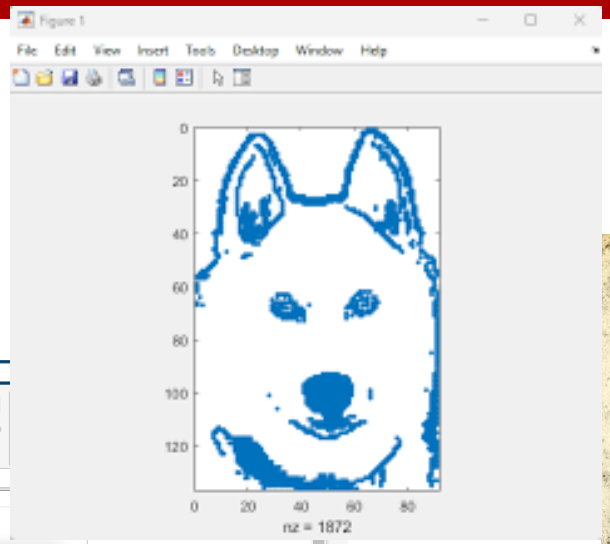

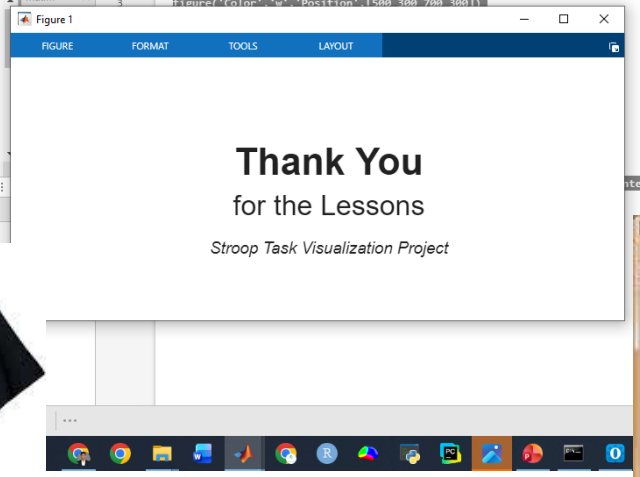

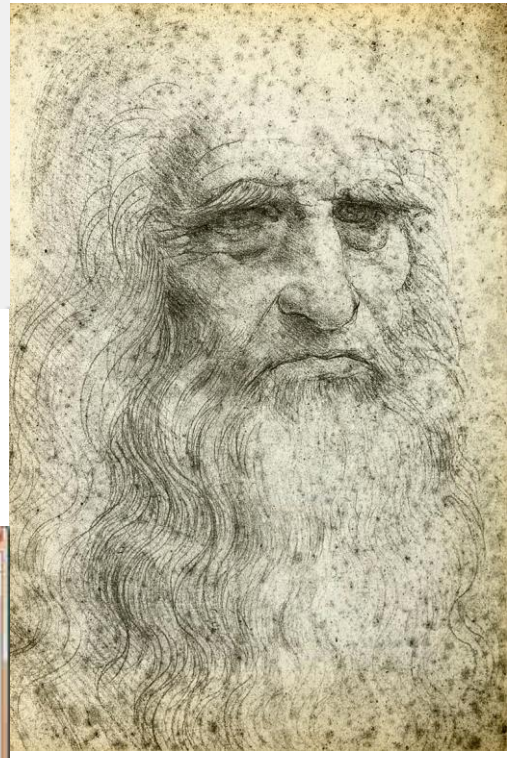

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